







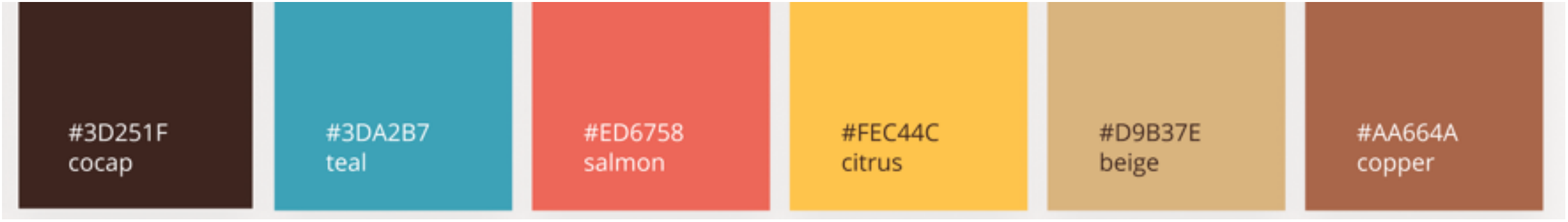


speed² of particle at location $x \sim$ kinetic energy

total ρ -weighted kinetic energy at time t

kinetic energy per unit volume transferred mass

total energy needed over total spent present



#3D251F
cocap

#3DA2B7
teal

#ED6758
salmon

#FEC44C
citrus

#D9B37E
beige

#AA664A
copper

Divergence of flux $\rho_t v_t \sim$ net flow of mass leaving/entering x



Local rate of change of density at x

Benamou–Brenier Formulation (Formal)

- Measures α, β , cost = $\|x - y\|_2^2$.

- Dynamic formulation of W_2 :

$$W_2^2(\alpha, \beta) = \inf_{\rho_t, v_t} \int_0^1 \int_{\mathcal{X}} \|v_t(x)\|^2 \rho_t(x) dx dt$$

kinetic energy per unit volume of transported mass

speed² of particle at location $x \sim$ kinetic energy

where ρ_t, v_t : measure, vector field at time t

total ρ -weighted kinetic energy at time t

total energy needed over transport process

- Subject to continuity equation:

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0, \quad \rho_0 = \alpha, \rho_1 = \beta$$

$\approx$ change in density caused only by only mass in/out-flows, not because it appears/disappears

Divergence of flux $\rho_t v_t \sim$ net flow of mass leaving/entering x

Local rate of change of density at x

Intuition: Benamou-Brenier

α



β

